SIMATS ENGINEERING

Name	Register Number	Course Code	Course
S Susmitha	192571068	ITA0510	Computer Vision

28. Find the boundary of the image using Convolution kernel for the given image.

AIM:
The boundary of the image using Convolution kernel for the given image.

PROGRAM:
```
import cv2
import numpy as np

img = cv2.imread(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png")

if img is None:
    print("Error: Image not found!")
else:
    # Sharpening kernel
    kernel = np.array([
        [-1, -1, -1],
        [-1,  9, -1],
        [-1, -1, -1]
    ])

    # Apply convolution
    sharpened_img = cv2.filter2D(img, -1, kernel)

    # Save the output image
    cv2.imwrite(
        r"C:\Users\Susmitha\Downloads\ITA0510 Lab\convolutional_image.jpg",
        sharpened_img
    )

    # Display the images
    cv2.imshow("Input Image", img)
    cv2.imshow("Convolutional Image", sharpened_img)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

INPUT:

OUTPUT:

RESULT:

Thus the program is executed successfully.